

Where does the image focus?

Refractive correction moves the focal point onto the retina.

Four patterns of refractive difficulty

Pattern	Optical problem	Correction / clue
Myopia	Distant focus in front of the retina	Concave lens; distance worse than near
Hyperopia	Unaccommodated focus behind the retina	Convex lens adds converging power
Astigmatism	Different powers in different meridians	Cylindrical correction
Presbyopia	Reduced near-focusing ability with age	Near addition; distance may remain clear

Apply the comparison

A longer globe or excessive refractive power can cause myopia; a shorter globe or insufficient power can cause hyperopia. Astigmatism is a meridional difference rather than simply too much or too little spherical power. Ask which task is blurred and compare distance with near acuity before choosing a correction.

The near response works together

Changing fixation from far to near links accommodation, convergence and miosis.

Coordinated changes

Accommodation

The lens provides extra refractive power to focus divergent rays from a near target.

Convergence

Both eyes turn inward to keep the near target aligned.

Miosis

Both pupils become smaller as part of the near response.

Why the examination target matters

A patient looking at the examiner's face or another close object may engage the near response during a pupil examination. Use a distant target when testing the light reflex so near focusing does not confuse interpretation. Similar-sized retinal images on near and distance charts do not remove the different focusing demands: a patient may read one chart well but struggle with the other.